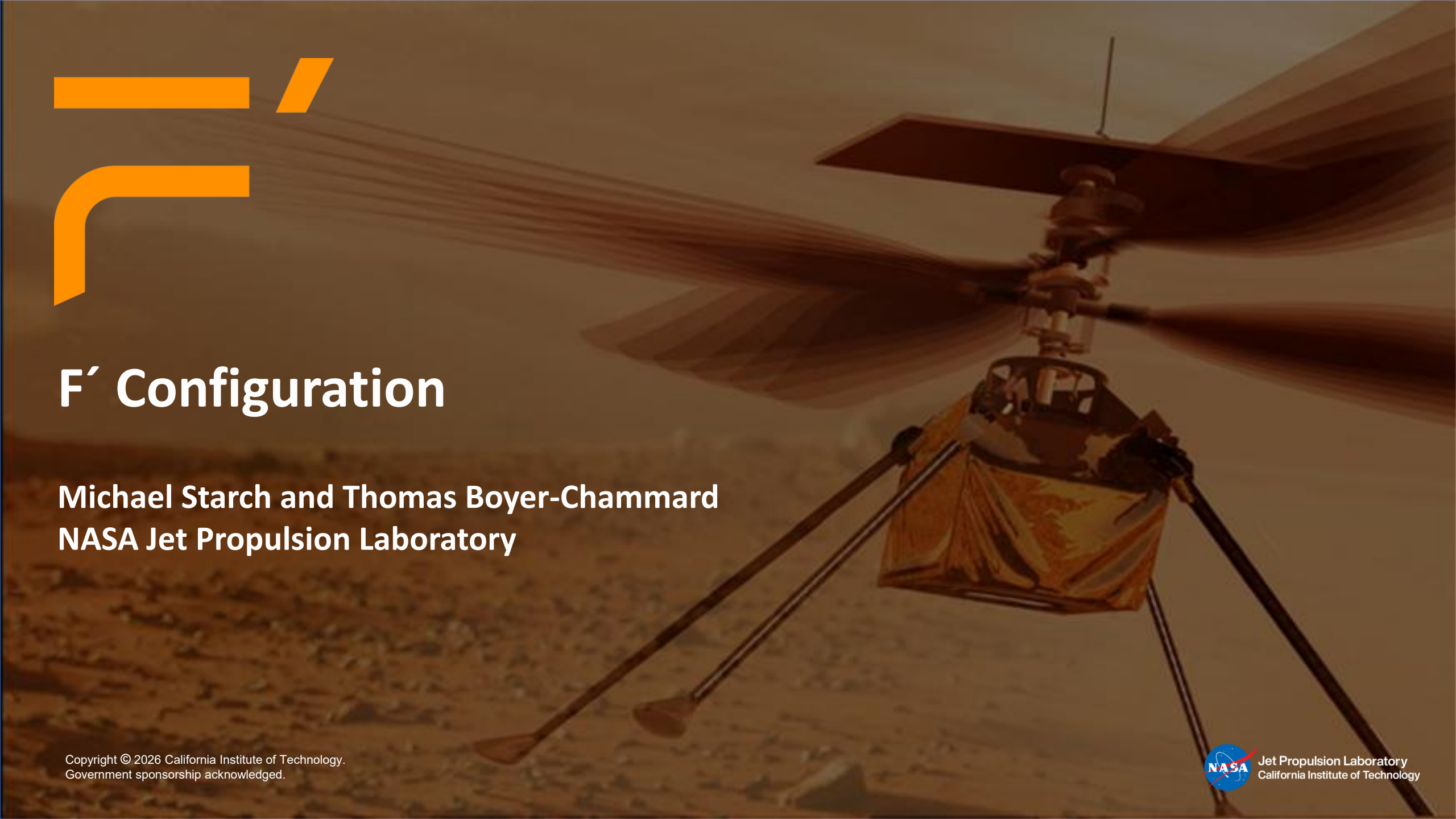


F´ Configuration

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Configuration Concepts

- Configuration source module: shipped default settings
- Configuration overrides: overlay of shipped default settings
  - Overrides are done on a per-file basis
  - Libraries may override creating new defaults
  - Projects perform final overlay
- Precedence is done by CMakeLists.txt ordering
  - Project > Libraries > F Prime
  - Define config override modules at top of CMakeLists.txt

- Project **must** understand and own configuration

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Sources of Configuration

- F Prime:
  - Framework configuration
  - Subtopology configuration
- F Prime Libraries
- Your Project!

- Available configuration is typically documented in SDDs

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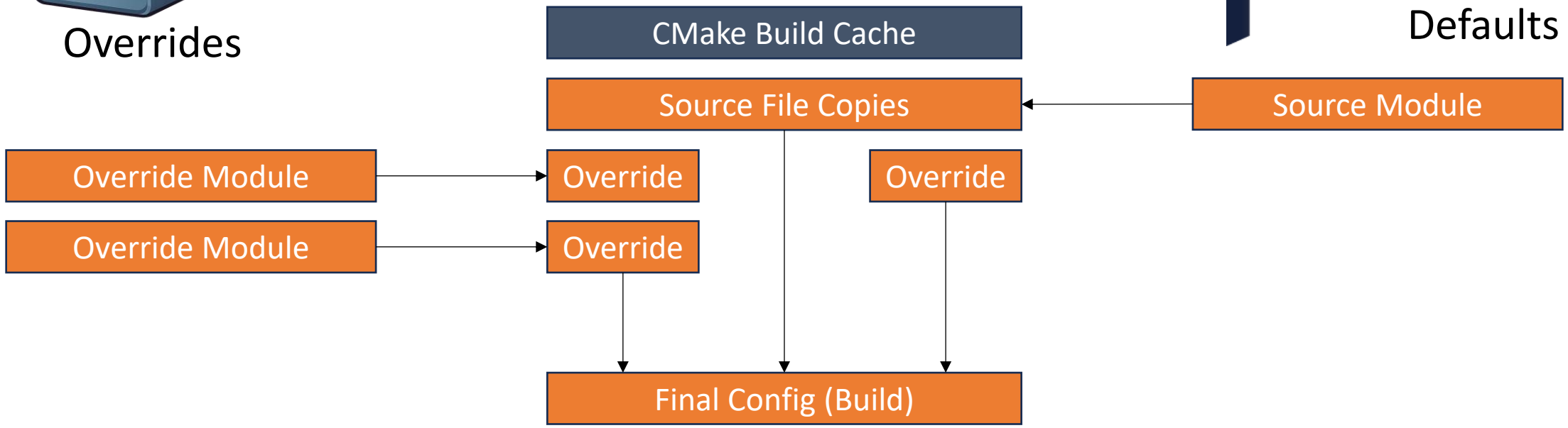
How Does Configuration Work



Overrides



Defaults



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Shared Flight / Ground Config

- Configuration values needed by the model are written to the dictionary
  - Must appear in FPP not header configurations
  - Determined by use within commands/events/telemetry etc.
- Use “dictionary” keyword to place other elements into the dictionary
- F´ GDS loads dictionary config and uses that to set type sizes and other properties

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Default Configuration

default/config

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Critical Config: Types

- F Prime has project defined types
  - FwSizeType, FwIndexType, Fw*Type
- Specified in FpConfig.fpp
- Intended to be set to fixed-width type
- E.g.
  - type FwSizeType = U64
- Caution: defaults are set to platform specified types

```
@ The unsigned type of larger sizes internal to the software,
@ e.g., memory buffer sizes, file sizes. Must be unsigned.
type FwSizeType = PlatformSizeType

@ The signed type of larger sizes internal to the software, used
@ for signed offsets, e.g., file seek offsets. Must be signed.
type FwSignedSizeType = PlatformSignedSizeType

@ The type of smaller indices internal to the software, used
@ for array indices, e.g., port indices. Must be signed.
type FwIndexType = PlatformIndexType

@ The type of arguments to assert functions.
type FwAssertArgType = PlatformAssertArgType

@ The type of task priorities used.
type FwTaskPriorityType = PlatformTaskPriorityType

@ The type of queue priorities used.
type FwQueuePriorityType = PlatformQueuePriorityType

@ The id type.
type FwIdType = U32
constant SIZE_OF_FwIdType = 4  @< Size of FwIdType in bytes

@ The type of task priorities used.
type FwTaskIdType = PlatformTaskIdType
```


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Critical Config: Constants

- Project configurable constants
  - Buffer sizes
  - String sizes
  - Table sizes
- Specified in FpConstants.fpp
- Tune F Prime for small platforms

```
@ When dumping the contents of the registry, this specifies the size of the buffer used to store object names.
@ Should be >= FW_OBJ_NAME_BUFFER_SIZE.
constant FW_OBJ_SIMPLE_REG_BUFF_SIZE = 255

@ Specifies the size of the string holding the queue name for queues
constant FW_QUEUE_NAME_BUFFER_SIZE = 80

@ Specifies the size of the string holding the task name for active components and tasks
constant FW_TASK_NAME_BUFFER_SIZE = 80

@ Specifies the size of the buffer that contains a communications packet
constant FW_COM_BUFFER_MAX_SIZE = 512

@ Specifies the size of the buffer attached to state machine signals
constant FW_SM_SIGNAL_BUFFER_MAX_SIZE = 128

@ Specifies the size of the buffer that contains the serialized command arguments
constant FW_CMD_ARG_BUFFER_MAX_SIZE = FW_COM_BUFFER_MAX_SIZE - SIZE_OF_FwOpcodeType - SIZE_OF_FwPacketDescriptorType

@ Specifies the maximum size of a string in a command argument
constant FW_CMD_STRING_MAX_SIZE = 40

@ Specifies the size of the buffer that contains the serialized log arguments
constant FW_LOG_BUFFER_MAX_SIZE = FW_COM_BUFFER_MAX_SIZE - SIZE_OF_FwEventIdType - SIZE_OF_FwPacketDescriptorType

@ Specifies the maximum size of a string in a log event
@ Note: This constant truncates file names in assertion failure event reports
constant FW_LOG_STRING_MAX_SIZE = 200

@ Specifies the size of the buffer that contains the serialized telemetry value
constant FW_TLM_BUFFER_MAX_SIZE = FW_COM_BUFFER_MAX_SIZE - SIZE_OF_FwChanIdType - SIZE_OF_FwPacketDescriptorType

@ Specifies the size of the buffer that contains the serialized telemetry value
constant FW_STATEMENT_ARG_BUFFER_MAX_SIZE = FW_CMD_ARG_BUFFER_MAX_SIZE
```


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Critical Config: Features

- F Prime has optional features
  - Object names
  - Registries
  - Tracing
- Specified in FpConfig.h
- Used to select/deselect features
  - 0 – Turn off
  - 1 – Turn on

```
24  // Allow objects to have names. Allocates storage for each instance
25  #ifndef FW_OBJECT_NAMES
26  #define FW_OBJECT_NAMES \
27      1  //!< Indicates whether or not object names are stored (more memory, ca
28  #endif
29
30  // To reduce binary size, FW_OPTIONAL_NAME(<string>) can be used to substitut
31  // when running with FW_OBJECT_NAMES disabled
32  #if FW_OBJECT_NAMES == 1
33  #define FW_OPTIONAL_NAME(name) name
34  #else
35  #define FW_OPTIONAL_NAME(name) ""
36  #endif
37
38  // Add methods to query an object about its name. Can be overridden by derive
39  // For FW_OBJECT_TO_STRING to work, FW_OBJECT_NAMES must be enabled
40  #if FW_OBJECT_NAMES == 1
41  #ifndef FW_OBJECT_TO_STRING
42  #define FW_OBJECT_TO_STRING \
43      1  //!< Indicates whether or not generated objects have toString() method
44  #endif
45  #else
46  #define FW_OBJECT_TO_STRING 0
47  #endif
48
49  // Adds the ability for all component related objects to register
50  // centrally.
51  #ifndef FW_OBJECT_REGISTRATION
52  #define FW_OBJECT_REGISTRATION \
53      1  //!< Indicates whether or not objects can register themselves (more co
54  #endif
55
56  #ifndef FW_QUEUE_REGISTRATION
57  #define FW_QUEUE_REGISTRATION 1  //!< Indicates whether or not queue registra
58  #endif
59
```


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Critical Config: Platform Config

- F Prime has configuration specified by platform
  - Handle sizes
  - OSAL configurations
- Specified in PlatformCfg.fpp
- Should be overridden by platform

- Project may override for multi-platform support

```
5   @ Maximum size of a handle for Os::Console
6   constant FW_CONSOLE_HANDLE_MAX_SIZE = 24
7
8   @ Maximum size of a handle for Os::Task
9   constant FW_TASK_HANDLE_MAX_SIZE = 40
10
11  @ Maximum size of a handle for Os::File
12  constant FW_FILE_HANDLE_MAX_SIZE = 16
13
14  @ Maximum size of a handle for Os::Mutex
15  constant FW_MUTEX_HANDLE_MAX_SIZE = 72
16
17  @ Maximum size of a handle for Os::Queue
18  constant FW_QUEUE_HANDLE_MAX_SIZE = 368
19
20  @ Maximum size of a handle for Os::Directory
21  constant FW_DIRECTORY_HANDLE_MAX_SIZE = 16
22
23  @ Maximum size of a handle for Os::FileSystem
24  constant FW_FILESYSTEM_HANDLE_MAX_SIZE = 16
25
26  @ Maximum size of a handle for Os::RawTime
27  constant FW_RAW_TIME_HANDLE_MAX_SIZE = 56
28
29  @ Maximum allowed serialization size for Os::RawTime objects
30  constant FW_RAW_TIME_SERIALIZATION_MAX_SIZE = 8
```


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Other Notable Configs

- AcConstants.fpp: general component config
- Specific Component Configuration
  - Many components have a config file for their properties
- VersionCfg.fpp
  - Allows custom project versions/library versions to be set up
- ComCfg.fpp
  - Configuration for ground interface stack

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Subtopology Configuration

Svc/Subtopologies/<Subtopology>/<Subtopology>Config/

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Subtopology Configuration

- Allows selection/injection of components
  - Example: FatalHandler
  - Example: Telemetry handler

- Allows tuning of subtopology properties
  - Base IDs
  - Task stack sizes and priorities
  - Queue sizes
- Strongly recommended when using a stock subtopology

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Critical Config: Component Selection

- Override component instance definition
- Must share “shape” of original
  - Typically done by importing interface
- Must be defined with the same name and parent module as original

```
module CdhCore{

    instance tlmSend: Svc.TlmChan base id CdhCoreConfig.BASE_ID + 0x06000 \
        queue size CdhCoreConfig.QueueSizes.tlmSend \
        stack size CdhCoreConfig.StackSizes.tlmSend \
        priority CdhCoreConfig.Priorities.tlmSend \

    # Uncommend the following block and comment the above block to use TlmPacketiz
    instance tlmSend: Svc.TlmPacketizer base id CdhCoreConfig.BASE_ID + 0x06000 \
        queue size CdhCoreConfig.QueueSizes.tlmSend \
        stack size CdhCoreConfig.StackSizes.tlmSend \
        priority CdhCoreConfig.Priorities.tlmSend \
    {
        # NOTE: The Name Ref is specific to the Reference deployment, Ref
        # This name will need to be updated if wishing to use this in a custom dep
        phase Fpp.ToCpp.Phases.configComponents """
        CdhCore::tlmSend.setPacketList(
            Ref::Ref_RefPacketsTlmPackets::packetList,
            Ref::Ref_RefPacketsTlmPackets::omittedChannels,
            1
        );
        """
    }
}
```


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Critical Config: Subtopology Properties

- Used to configure basic properties
  - Subtopology base ID
  - Task priority
  - Task stack size
  - Queue depth
- Can contain other properties
  - Example: Buffer manager bin sizes
- Must take the whole file

```
module CdhCoreConfig {
    #Base ID for the CdhCore Subtopology, all compor
    constant BASE_ID = 0x01000000

    module QueueSizes {
        constant cmdDisp     = 10
        constant events      = 10
        constant tlmSend     = 10
        constant $health     = 25
    }


    module StackSizes {
        constant cmdDisp     = 64 * 1024
        constant events      = 64 * 1024
        constant tlmSend     = 64 * 1024
    }

    module Priorities {
        constant cmdDisp     = 35
        constant $health     = 24
        constant events      = 23
        constant tlmSend     = 22

    }
}

```


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Conclusion

F´ Configuration Pitfalls and Solutions

- Module order is important
  - Place configuration and override modules first in a CMakeLists.txt
  - Define configuration modules at level common to all users
  - Set up order: F Prime --> then libraries --> then your project
- Configuration is explicit
  - Enumerate all CONFIGURATION_OVERRIDES
  - Ensure exact filename matches
- Configuration source namespacing folders are unique
  - ”config” is used by F-Prime
  - Choose ”<Library>Config” or “<Project>Config” for custom config folders
  - Use ‘#include “config/…”’, “#include “<Library>Config/…” etc.
- Updates can change configuration
  - Take care to release notes for file name changes, new settings, etc.

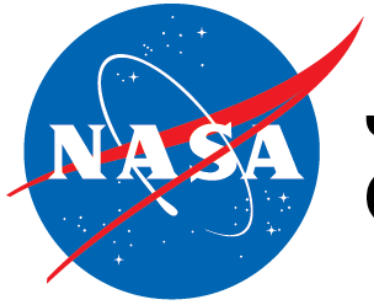
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Takeaways

- Project may use defaults, but must own all configuration
- Study and understand all configurations
- Be cautious of updates, read all release notes

- Configuration is yours, enjoy!

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